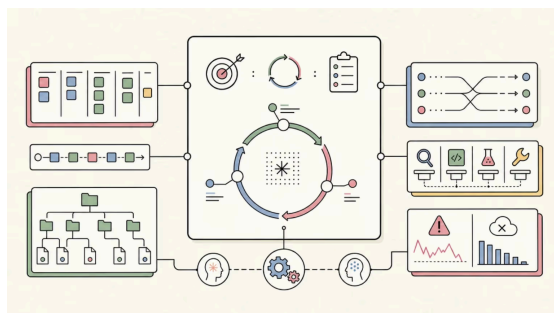


How Quants Use Loop Engineering to Find Trades That Actually Work (The Full Framework)

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The reason most trading strategies look incredible on a backtest and then lose money the moment you trade them live is because they were tested once on the same data they were built on. You came up with an idea, tested it on historical data, the chart looked great, you started trading, and two weeks later you are bleeding money wondering what went wrong.

What went wrong is that you found a pattern in the noise and mistook it for a real edge. The backtest was not testing your strategy. It was confirming your bias.

Quants do not work like this. They run a loop. Generate a strategy, test it, score it against a specific metric, read why it failed, feed that information back in, and generate a better version. Then they repeat. Each round keeps the variants that scored well and kills the ones that did not. And at the end every surviving strategy has to pass one final gate on data it has never seen before. If it does not hold up on fresh data it gets thrown out no matter how good it looked before.

This guide walks you through the exact loop framework with the scoring metrics, the stop rules, and the out-of-sample gate that separates real edges from faster overfitting.

WHY SINGLE-PASS TESTING ALWAYS FAILS

When you test a strategy once on historical data you are running a closed-book exam where the student already saw the answer key. The strategy "learned" from the data and then got tested on the same data it learned from. Of course it looks good. That does not mean it will work tomorrow.

The deeper problem is that financial data is noisy. There are thousands of patterns in any dataset and most of them are random. If you look hard enough you will always find something that worked in the past. The question is whether it worked because of a real structural reason or because of coincidence.

A single test cannot answer that question. A loop can because it generates dozens of variations, scores them rigorously, and forces the survivors through a gate that eliminates luck.

THE LOOP: HOW IT WORKS

The loop has 5 stages that repeat. Each round takes the output from the previous round and makes it better.

Stage 1 is hypothesis generation. You start with a trading idea. It can be as simple as "buy when the RSI drops below 30 on the daily chart" or as complex as "combine momentum on the weekly timeframe with mean reversion on the hourly timeframe and filter by volatility regime." The key is that you do not need the idea to be perfect. You just need a starting point because the loop refines it.

If you are using Claude Code or Claude Fable to generate strategies, give it a prompt like this:

"Generate 10 mean reversion trading strategies for the S&P 500 using 15-minute candles. Each strategy should use a different combination of indicators and entry /exit rules. Include the exact parameters for each indicator. Format each strategy as: entry condition, exit condition, stop loss rule, and position sizing rule."

Stage 2 is backtesting. You run each strategy against historical data and record the results. Not just whether it made money but the specific metrics that tell you if the edge is real or just noise.

Stage 3 is scoring. You score each strategy using a metric called the Information Coefficient Ratio which I will explain in detail in the next section. This score tells you whether the strategy is finding real signal or just getting lucky on random patterns.

Stage 4 is analysis and feedback. You read why the losing strategies failed. What market conditions broke them. What assumptions were wrong. What edge cases they missed. This analysis gets fed back into Stage 1 as constraints for the next round. "Generate 10 more strategies but avoid the failure modes we found in round 1."

Stage 5 is selection. You keep the strategies that scored above your threshold and discard the rest. The survivors go into the next round where the loop tightens further.

After 3 to 5 rounds of this loop you have a small set of strategies that survived multiple rounds of testing and refinement. These are your candidates for the final gate.

THE SCORING FUNCTION: IC AND ICIR

The loop only works if you have a scoring function that can tell the difference between a real edge and noise. Without it the loop optimizes for nothing and you are just generating random strategies with no way to rank them.

The standard metric quants use is called the Information Coefficient or IC. It measures the correlation between what your strategy predicted would happen and what actually happened. If your signal said "this stock will go up" and it went up, that is a positive IC. If it said "up" and it went down, that is negative.

A single IC reading is noisy though. A strategy could have a great IC one month and a terrible IC the next. What actually matters is how consistent the IC is over time. That consistency is measured by ICIR which stands for Information Coefficient to Information Ratio. The formula is simple:

$$\text{ICIR} = \text{mean(IC)} / \text{standard deviation(IC)}$$

A strategy with a modest but steady IC of 0.03 every month is far more valuable than a strategy with an IC of 0.15 one month and negative 0.10 the next. The first one is a real edge you can trade. The second one is noise that happened to work sometimes.

Here is how to interpret ICIR scores:

ICIR above 0.5: Strong. This is a genuinely consistent signal worth trading. ICIR between 0.3 and 0.5: Moderate. Worth investigating further but needs more validation. ICIR below 0.3: Weak. Probably noise. The loop should deprioritize strategies scoring here.

To calculate IC and ICIR for your strategies, give Claude this prompt:

"I have backtest results for 10 trading strategies tested over 12 months of daily data. For each strategy I have the signal value it generated each day and the actual return that followed. Calculate the Information Coefficient for each month (the Pearson correlation between the signal values and the subsequent returns). Then calculate the ICIR for each strategy (mean IC divided by standard deviation of IC across all months). Rank the

strategies by ICIR highest to lowest. Flag any strategy with ICIR below 0.3 as likely noise."

THE DECAY CHECK: IS THE EDGE STILL ALIVE

Even a real edge has a shelf life. Markets change. Other participants find the same pattern. Regulations shift. What worked 6 months ago might be dead today.

You measure this by looking at how quickly the signal's predictive power decays over time. The technical term is the half-life of the signal.

A signal with a half-life of 50 days means it takes about 50 days for the signal's predictive power to drop to half its original strength. That is a tradeable edge because you have plenty of time to act on it before it fades.

A signal with a half-life of 2 days means the predictive power is basically gone by the time you execute the trade. You are paying transaction costs to chase something that already disappeared. That is not tradeable for most people.

To check the decay of your surviving strategies, give Claude this prompt:

"For each of my surviving strategies, calculate the autocorrelation of the signal at lags of 1 day, 5 days, 10 days, 20 days, and 50 days. Plot the decay curve showing how the signal's predictive power drops over time. Estimate the half-life of each signal. Flag any strategy where the half-life is under 5 days as likely too short to trade profitably after transaction costs."

The loop should automatically reject strategies with a half-life shorter than your trading frequency. If you are making trades that take 3 to 5 days to play out, a signal that decays in 2 days is useless even if its ICIR is strong.

THE OUT-OF-SAMPLE GATE

This is the most important step in the entire framework and the one that most people skip. Everything up to this point was done on the same historical data. The loop refined strategies. The scoring function ranked them. The decay check filtered out the short-lived ones. But all of that happened on data the strategies have already seen.

The out-of-sample gate tests every surviving strategy on completely fresh data that was held back from the beginning and never used at any point during the loop. This is data the strategies have never seen and could not have been optimized against.

The practical way to do this is to split your historical data before you start the loop. Take the most recent 20 to 30% of your data and set it aside. Lock it away. Run the entire loop on the remaining 70 to 80%. Only after the loop is finished and you have your surviving strategies do you test them on the held-out data.

The rules for the gate are strict:

The ICIR must hold on the out-of-sample data. If a strategy scored 0.6 ICIR on the in-sample data but drops to 0.15 on the out-of-sample data, it was overfit. Throw it out.

The decay profile must hold. If the half-life was 30 days in-sample but drops to 3 days out-of-sample, the signal was not real. Throw it out.

Adjust for multiple testing. This is the part most people get wrong and it is critical. If you tested 200 strategies inside the loop and one of them passes the out-of-sample gate, that could still be luck. When you test 200 things, some will pass by chance alone. You have to raise the bar based on how many attempts you made.

The standard correction is the Bonferroni adjustment where you divide your significance threshold by the number of strategies tested. If your original threshold was a p-value of 0.05 and you tested 200 strategies, your adjusted threshold is 0.05 divided by 200 which equals 0.00025. Only strategies that pass at this stricter level are considered genuinely significant.

To run the out-of-sample gate, give Claude this prompt:

"I have [X] strategies that survived the loop. I held back 20% of my data as an out-of-sample set that was never used during optimization. For each surviving strategy, calculate the ICIR on the out-of-sample data. Also calculate the signal half-life on the out-of-sample data. Compare both metrics to the in-sample values. Flag any strategy where the out-of-sample ICIR drops by more than 50% from the in-sample ICIR. Apply a Bonferroni correction for [X] strategies tested and flag any that do not pass the adjusted significance threshold. Only strategies that pass all three checks (ICIR holds, decay holds, significance after correction) should be marked as viable."

PUTTING IT ALL TOGETHER: THE FULL WORKFLOW

Step 1: Split your historical data. Hold back the most recent 20 to 30% as your out-of-sample set. Do not touch it until the very end.

Step 2: Generate your first batch of strategies. Start with 10 to 20 variations of a core idea.

Step 3: Backtest all of them on the in-sample data.

Step 4: Score each one using ICIR. Kill anything below 0.3.

Step 5: Check the decay half-life. Kill anything under 5 days.

Step 6: Analyze why the losers failed. What market conditions, what assumptions, what edge cases.

Step 7: Feed the failure analysis back into generation. Create 10 to 20 new strategies that avoid the failure modes from round 1.

Step 8: Repeat steps 3 through 7 for 3 to 5 rounds total. Each round the surviving pool gets smaller and stronger.

Step 9: Take all final survivors and test them on your held-out out-of-sample data. Apply the Bonferroni correction for the total number of strategies tested across all rounds.

Step 10: Only strategies that pass the out-of-sample gate with corrected significance are real. Everything else gets discarded.

This entire workflow can run in Claude Code or Claude Fable. You describe the strategy ideas, Claude generates the code, runs the backtests, calculates the metrics, and reports the results. The loop itself is the human directing each round based on what the previous round revealed.

WHAT THE LOOP SHOULD NEVER DO

Optimize entry and exit parameters to fit historical data perfectly. If your strategy worked because you used a 17-period moving average instead of a 20-period one, that specificity is almost certainly overfit. Real edges work across a range of reasonable parameters, not just one magic number.

Test on data that overlaps with the out-of-sample set. If even one data point from the held-out set leaks into the loop, the gate is compromised.

Run indefinitely without a round cap. Set a maximum of 5 rounds. More rounds means more opportunity to accidentally overfit even with the scoring function in place.

Skip the multiple testing correction. The more strategies you test, the higher your bar needs to be. 200 strategies tested with no correction is just automated luck-finding.

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IMPORTANT DISCLAIMERS

This guide is for educational purposes only. Trading involves real financial risk. Past performance does not guarantee future results. Even strategies that pass the full loop and out-of-sample gate can lose money in live markets due to execution slippage, changing market conditions, or structural regime shifts.

The framework described here is a simplified version of what professional quantitative trading firms use. Real firms layer additional checks including walk-forward analysis, Monte Carlo simulation, and live paper trading periods before committing real capital.

Always paper trade any strategy for a minimum of 4 to 8 weeks before using real money. Start with position sizes you can afford to lose entirely. Never trade with borrowed money.